

Practice Quiz: Perception (Implementation)

Name: __ Date: __

Part A: Multiple Choice

1. Variational message passing differs from simple Bayesian updating because: A) It doesn't use Bayes' theorem B) It iterates, combining messages from past, present, and future until convergence C) It's slower and less accurate D) It only works with continuous states
2. The three messages combined at each time step are: A) Input, output, error B) Observation likelihood, prior/transition from past, and message from future C) Left, right, center D) Fast, medium, slow
3. Prediction error in this implementation is computed as: A) The sum of all observations B) The difference between actual and predicted observations (or states) C) A random number D) The log of the A matrix
4. Precision weighting multiplies prediction errors by: A) Zero B) A scalar precision value — higher precision means more influence C) The number of states D) Negative one
5. Free energy $F = \text{Complexity} - \text{Accuracy}$ means: A) Simpler models are always better B) Good inference balances explaining data (accuracy) and staying close to priors (complexity) C) Complexity is always bad D) Accuracy doesn't matter
6. Working in log space (log probabilities) is important because: A) Logs are easier to type B) It prevents numerical underflow from multiplying many small probabilities C) Python requires it D) It makes values negative
7. The convergence of message passing means: A) The algorithm crashes B) Beliefs stop changing significantly between iterations C) All states become equally likely D) Observations stop arriving

Part B: Short Answer

1. Trace through one iteration of the message passing algorithm for $t=1$ in a 2-state model. Show the three messages being combined and the resulting belief update.
2. Explain how precision weighting implements attention in code. Give an example where you would set high precision for one observation channel and low for another.
3. Design a test to verify your free energy computation is correct. What properties should F have? (e.g., F should decrease when beliefs improve, F should decompose correctly into accuracy and complexity)